Bitcoin Trader Analysis using Market Sentiment

Data Science Assignment Report

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1. Introduction

The cryptocurrency market is highly dynamic and influenced by investor sentiment. The Bitcoin Fear & Greed Index reflects overall market psychology, while historical trading data provides valuable insights into trader behavior and performance. This project combines both datasets to analyze how market sentiment affects trading outcomes. Through **data preprocessing, exploratory data analysis (EDA), and visualization**, the project aims to identify meaningful patterns and generate actionable insights for better trading decisions.

2.Objective

- To analyze historical Bitcoin trader performance using trading data.
- To integrate trader data with the Bitcoin Fear & Greed Index based on the trading date.
- To perform data cleaning and preprocessing for accurate analysis.
- To explore how market sentiment influences trader profitability and trading behavior. .
- To generate actionable insights and recommendations that can help improve trading decisions and risk management.

3.Dataset Description

1.Historical Trader Data: Contains account details, execution price, leverage, size, side, closedPnL, etc.

Rows : 211224, **Columns** : 16

2.Fear & Greed Index: Contains Date and Market Sentiment Classification.

Rows : 2644, **Columns** : 4

4.Data Cleaning & Preprocessing

1.Missing Values

Objective: To identify missing values that could affect the analysis.

Method: The dataset was checked using **isnull().sum().**

Result: No missing values were found in any of the columns. Therefore, no missing value treatment was required**.**

2.Duplicate Records

Objective: To identify duplicate entries that could bias the analysis.

Method: Duplicate records were checked using **duplicated().sum().**

Result: No duplicate records were found in the dataset. Hence, no duplicate removal was necessary.

3.Data Type Verification

Objective: To ensure each column has the appropriate data type.

Result:

- Converted the **Timestamp IST** column to datetime format using the specified date-time pattern.
- Converted the **Timestamp** column to datetime format.
- Converted the **timestamp** and **date** columns in the Fear & Greed dataset to datetime format.
- Extracted the date from **Timestamp IST** to create a common **date** column for merging both datasets.

4.Dataset Merge

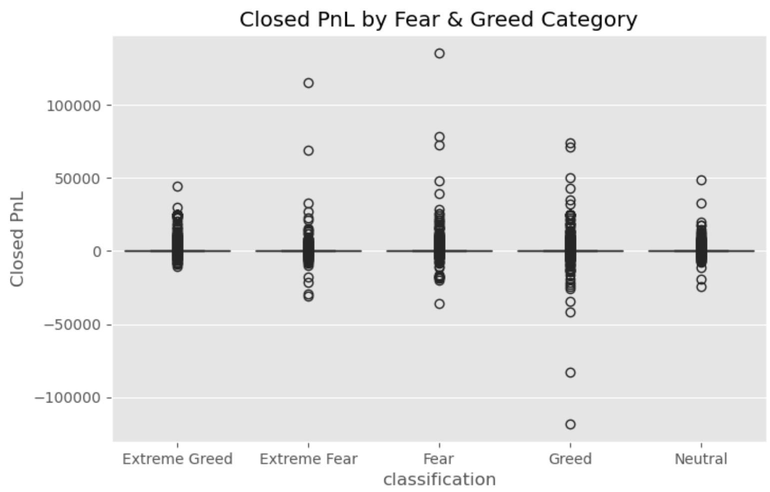
Objective: To combine historical trader data with the Fear & Greed Index.

Result: The datasets were successfully merged using the **Date** column, producing a unified dataset for further analysis.

5. Exploratory Data Analysis(EDA)

1. Profit/Loss by Market Sentiment

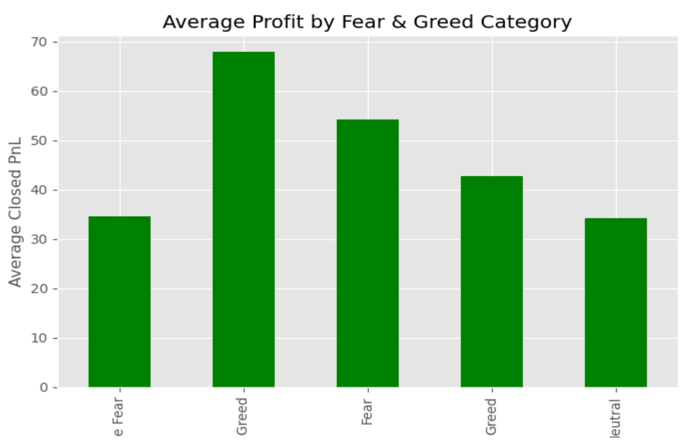
Objective: To analyze how trader profitability (Closed PnL) varies across different Fear & Greed market categories.



Observation: The box plot shows how the distribution of Closed PnL differs across Fear & Greed market categories. It highlights the median, spread, and outliers for each category.

2. Average Profit by Sentiment

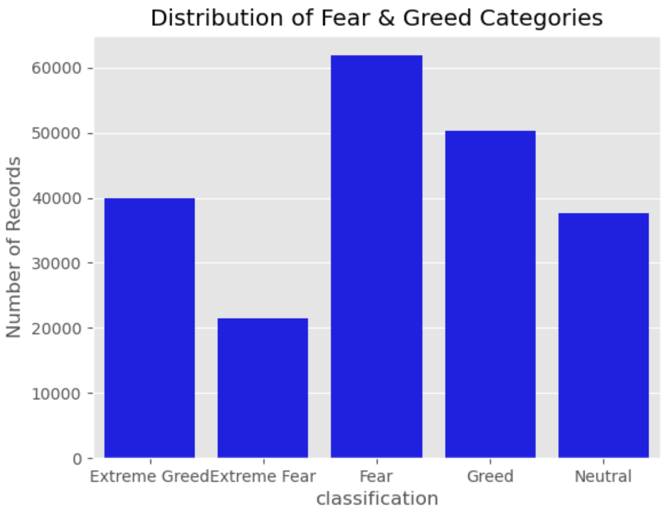
Objective: To compare the average trader profit (Closed PnL) across different Fear & Greed market categories.



Observation: The bar chart compares the average Closed PnL for each Fear & Greed market category. Differences in bar heights indicate how average trader profitability varies under different market conditions.

3. Fear & Greed Categories Distribution

Objective: To analyze how frequently each Fear & Greed market category appears in the dataset.



Observation: This graph helps understand the overall market behavior distribution. If "Fear" or "Greed" dominates, it shows that the market spends most of its time in that condition. Traders can use this information to understand typical market environments and adjust their strategies accordingly.

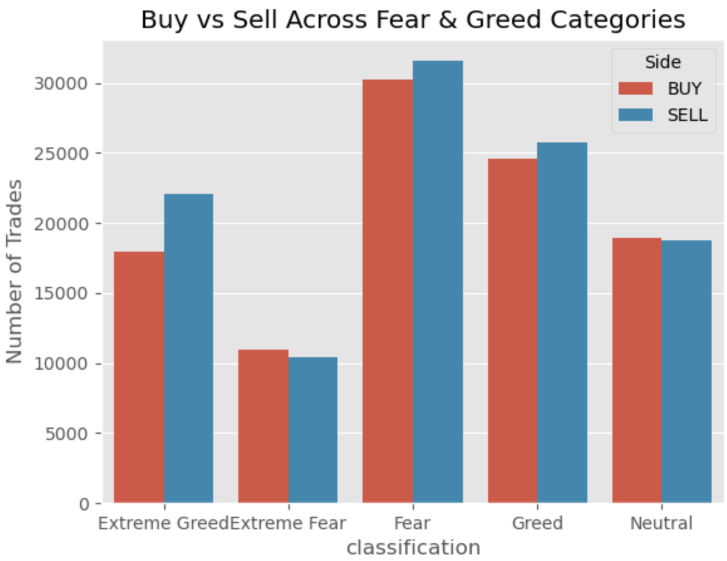
4. Trading Volume by Fear & Greed Category

Objective: To analyze how total trading volume (Size USD) varies across different Fear & Greed market categorie

Observation: The trading volume is higher during certain market conditions, especially during Greed or Extreme Greed phases, while lower volumes are observed during Fear periods.

5. Buy vs Sell Trades across Fear & Greed Categories

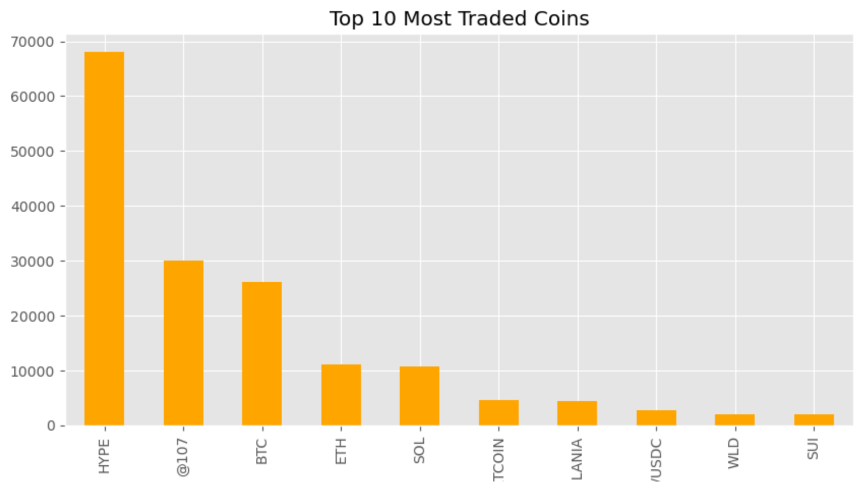
Objective: To analyze how trader behavior (Buy vs Sell trades) changes under different Fear & Greed market conditions.



Observation: Trader behavior is influenced by market sentiment. During Fear conditions, selling activity tends to increase as traders reduce exposure, while during Greed conditions, buying activity dominates as traders show higher risk appetite. This indicates that market psychology directly impacts trading decisions.

6. Top 10 Most Traded Coins

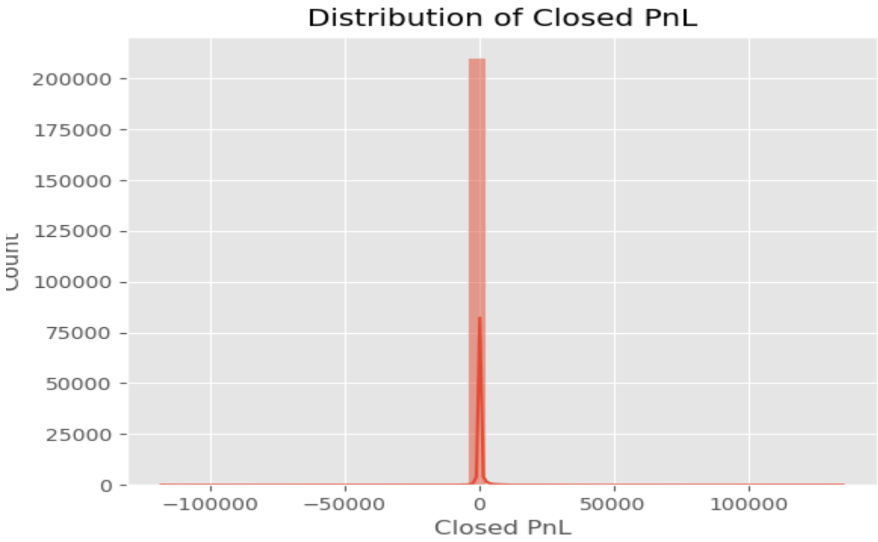
Objective: To identify the top 10 most frequently traded cryptocurrencies in the dataset.



Observation: The graph shows that a few coins dominate trading activity, while the rest have significantly lower trade counts.

7. Profit Distribution

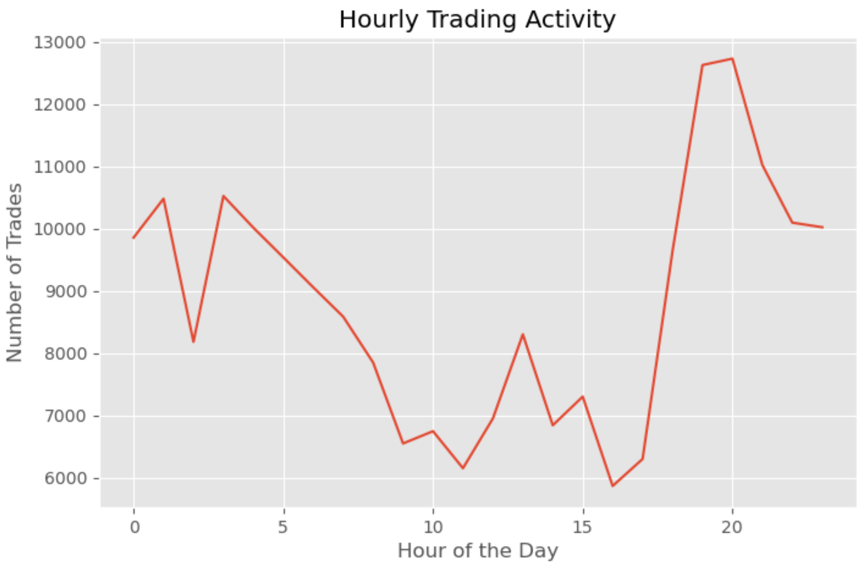
Objective: To analyze how trader profits and losses (Closed PnL) are distributed across all trades



Observation: Most trades are clustered around small profit or loss values, while only a few trades show very high profits or large losses, indicating the presence of outliers in the dataset.

8. Hourly Trading Activity

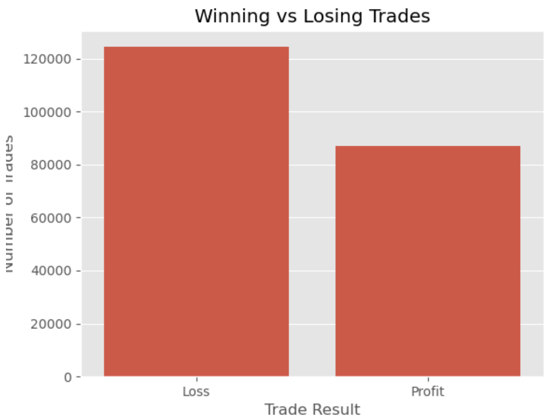
Objective: To analyze at which hours of the day trading activity is highest.



Observation: This helps identify peak trading hours, which can be useful for understanding trader behavior and market activity patterns.

9. Winning Vs Losing Trades

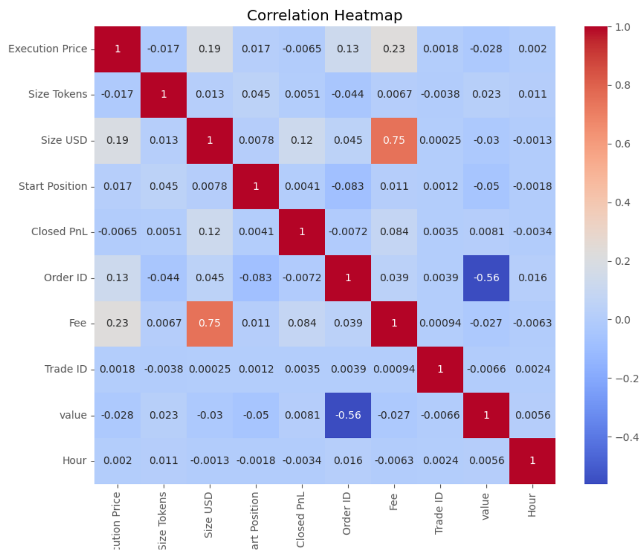
Objective: To analyze the distribution of profitable and loss-making trades.



Observation: The graph shows the number of profitable trades (Profit) and loss-making trades (Loss). It helps identify whether traders are generally making more profits or losses.

10. Correlation Heatmap

Objective: To understand the relationship between numerical variables such as trade size, profit/loss, fees, and execution price.



Observation: The heatmap shows the correlation between different numerical features. Some variables like Size USD and Size Tokens show strong positive correlation, while Closed PnL has weak correlation with most features.

7.Key Insights

- Fear was the most common market sentiment with 61,837 trades, showing that traders were most active during uncertain market conditions.
- Extreme Greed had the highest average profit (Closed PnL = 67.89), indicating that traders earned better average returns during strong bullish market sentiment.
- Fear recorded the highest trading volume (about 483.3 million USD), which means the largest amount of trading activity happened during Fear market conditions.
- The dataset contained more losing trades (124,355) than profitable trades (86,869), showing that making consistent profits in cryptocurrency trading is challenging.
- Trading was mainly concentrated in a few cryptocurrencies. HYPE was the most traded coin (68,005 trades), followed by @107 and BTC.
- The analysis shows that market sentiment affects trading activity and profitability, but successful trading also depends on good strategy and proper risk management.

8.Recommendations

- Use the Fear & Greed Index together with technical analysis instead of relying on market sentiment alone.
- Apply proper risk management because the dataset shows that losing trades are more common than profitable trades.
- Focus on high-volume cryptocurrencies and regularly analyze past trading performance to improve future trading decisions.

9.Conclusion

This project explored the relationship between Bitcoin market sentiment and trader performance by integrating historical trading data with the Fear & Greed Index. Through data cleaning, preprocessing, dataset merging, and exploratory data analysis, several meaningful patterns were identified. The analysis showed that Fear market conditions had the highest trading activity, while Extreme Greed recorded the highest average trader profitability. It also revealed that loss-making trades were more frequent than profitable ones, emphasizing the importance of disciplined trading and effective risk management. Overall, the findings demonstrate that market sentiment influences trading behavior and performance, while also providing valuable insights that can support more informed and data-driven trading decisions.